

Lecture 9

Boundary Value Problems

ASDA/Dynamics of Rotor Systems

Brano Titurus

QB 2.64 UoB

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This lecture

- Static BVPs
- Harmonic BVPs
- Eigenvalue BVPs

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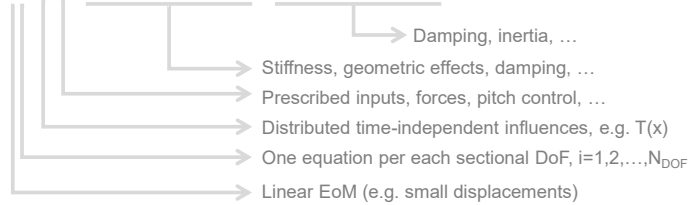


General problem description

Rotor blade (and other slender lifting surface) equations are modelled using 1D space-time PDEs (field equations) with the corresponding BCs and ICs.

Field equations:

$$g_i(x, t, w(x, t), w'(x, t), \dots, w(x, t), \dot{w}(x, t), \dots) = 0 \quad i = 1, \dots, N_{dof}$$



Boundary conditions:

$$h_j(t, w(x_\beta, t), w'(x_\beta, t), \dots, w(x_\beta, t), \dot{w}(x_\beta, t), \dots) = 0 \quad j = 1, \dots, N_{ord}$$

Initial conditions:

$$w(x, 0) = w_0(x), \dot{w}(x, 0) = \dot{w}_0(x)$$

Total order of the system of PDEs.

Domain of definition:

$$t \geq 0, x \in [x_r, x_t], x_t = R, \beta = \{r, R\}$$

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Boundary Value Problems

Different types of BVPs can be constructed using original IV-BVP depending on further problem task specialization:

- Static BVP
- Harmonic BVP
- Eigenvalue BVP

Static BVP describes the time-independent static or quasi-static structural response (all time derivative terms and time dependent functions are zero). Resulting function describes the static deflection shape.

Harmonic BVP describes the steady state harmonic response caused by the applied (distributed, concentrated) harmonic loads with known excitation frequency. Resulting function describes the response amplitudes.

Eigenvalue BVP describes the free (natural) harmonic response of structure. Resulting function (eigenfunction) describes the response amplitude shape and resulting parameter (eigenvalue) is the square of the vibration frequency.

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Types of (flapping beam) BVPs

Consider the field equation of the rotating flapping blade:

$$(EI_F w'')'' - (T w')' + m\ddot{w} = L \quad w(x,t) = ?$$

Static BVP: all time derivative components set to zero:

$$(EI_F w'')'' - (T w')' = L \quad w(x) = ?$$

Harmonic BVP: lift $L(x,t) = L_0 e^{i\omega t}$, assumed response $w(x,t) = W(x) e^{i\omega t}$

Type 1

$$(EI_F W'')'' - (T W')' - \omega^2 m W = L_0 \quad W(x) = ?$$

Harmonic BVP: boundary load $F(x_\beta, t) = F_0 e^{i\omega t}$, assumed response $w(x,t) = W(x) e^{i\omega t}$

Type 2

$$(EI_F W'')'' - (T W')' - \omega^2 m W = 0 \quad W(x) = ?$$

Eigenvalue BVP: assumed free and natural response $w(x,t) = W_j(x) e^{i\omega_j t}$

$$(EI_F W_j'')'' - (T W_j')' - \omega_j^2 m W_j = 0 \quad \omega_j, W_j(x) = ?$$

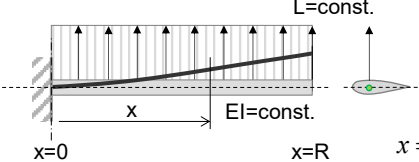
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Solving static BVP

The simplest BVP is the static response problem. By default, the time component is not present in these formulations, they are time-independent.

Let's consider a simple bending problem, e.g. non-rotating blade with the fixed root and constant distributed lift:



$$\begin{array}{ll} x=0, w(0)=0 & x=R, w''(R)=0 \\ x=0, w'(0)=0 & x=R, w'''(R)=0 \end{array}$$

Bending equation for this case:

$$EI w^{iv}(x) = L$$

After four integrations:

$$w(x) = (L/24EI)x^4 + (C_1/6)x^3 + (C_2/2)x^2 + C_3x + C_4$$

Use four BCs to find $C_1, \dots, C_4$; use constitutive relationship to determine M and S BCs.

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Formulating harmonic BVP

Consider coupled bending-twist of a non-rotating blade with fixed root, distributed loads $L(x,t)=L_0 e^{i\omega t}$, $M_A=0$ and vertical tip load applied at the shear center $F(x_{tip},t)=F_0 e^{i\omega t}$. Write down the corresponding harmonic BVP.

The original IV-BVP field equations:

$$(EI w'')'' + m \ddot{w} + me \ddot{\theta} = L$$

$$(GJ \theta')' - I_{e.c.} \ddot{\theta} - me \ddot{w} = -(L e_A + M_A)$$

Assumed harmonic responses at known ω : $w(x,t) = W(x) e^{i\omega t}$, $\theta(x,t) = \Theta(x) e^{i\omega t}$

$$w'' = W'' e^{i\omega t}, \theta'' = \Theta'' e^{i\omega t}, \dot{w} = -\omega^2 W e^{i\omega t}, \ddot{\theta} = -\omega^2 \Theta e^{i\omega t}$$

Field equation after substituting L , M_A , w , θ :

$$(EI W'')'' - \omega^2 m(W + e\Theta) = L_0$$

$$(GJ \Theta')' + \omega^2 (I_{e.c.} \Theta - meW) = -L_0 e_A$$

ω is known and
 $W(x)$ and $\Theta(x)$ are the unknowns.

Time-independent boundary conditions after substituting w , θ :

$$x = x_{root} : W = 0, W' = 0, \Theta = 0$$

$$x = x_{tip} : M = 0, S = F_0, Q = 0$$

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Field equation for flapping blade eigenvalue BVP

Derive the BVP field equation for rotating flapping blade with constant EI_F and m , blade root with the flapping spring k_F and free tip. The rotor radius is R and rotor speed is Ω .

Consider free flapping blade, i.e. $L=0$. Assume free response $w(t,x)$ as follows:

$$w(t,x) = W(x) \sin(\omega_F t)$$

Time and space derivatives, distributed axial force $T(x)$:

$$w'' = W'' \sin(\omega_F t), \ddot{w} = -\omega_F^2 W \sin(\omega_F t), \dots \quad T(x) = \Omega^2 \int_x^R m(\tilde{x}) \tilde{x} d\tilde{x} = \frac{\Omega^2 m}{2} (R^2 - x^2)$$

Substitute assumed response to the EoM:

$$(EI_F W'')'' \sin(\omega_F t) - (T W')' \sin(\omega_F t) - \omega_F^2 m W \sin(\omega_F t) = 0$$

This equation is valid for arbitrary time instant when:

$$(EI_F(x) W''(x))'' - (T(x) W'(x))' - \omega_F^2 m(x) W(x) = 0 \quad W(x), \omega_F = ?$$

Observe this important property –
the frequency is not known a priori.

Substituting $T(x)$ and completing all derivatives gives the final field equation:

$$EI_F W^{iv} - \frac{\Omega^2 m}{2} (R^2 - x^2) W'' + \Omega^2 m x W' - \omega_F^2 m W = 0$$

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Full flapping blade eigenvalue BVP

Use previously developed BCs and substitute the assumed solution:

$$x = e_F R : W(e_F R) \sin(\omega_F t) = 0$$

$$x = e_F R : W''(e_F R) \sin(\omega_F t) - (k_F / EI_F) W'(e_F R) \sin(\omega_F t) = 0$$

$$x = R : W''(R) \sin(\omega_F t) = 0$$

$$x = R : W'''(R) \sin(\omega_F t) = 0$$

Complete BVP definition:

$$EI_F W^{iv} - \frac{\Omega^2 m}{2} (R^2 - x^2) W'' + \Omega^2 m x W' - \omega_F^2 m W = 0 \quad e_F R < x < R \quad \text{Field equation}$$

$$\begin{bmatrix} W(e_F R) \\ W''(e_F R) - (k_F / EI_F) W'(e_F R) \\ W''(R) \\ W'''(R) \end{bmatrix} = \mathbf{0} \quad \text{Boundary conditions}$$

This problem has more than one *solution pairs* – the modes of vibration:

$$\underbrace{\omega_{F,1}, W_1(x)}_{\text{mode 1}}; \underbrace{\omega_{F,2}, W_2(x)}_{\text{mode 2}}; \underbrace{\omega_{F,3}, W_3(x)}_{\text{mode 3}}; \dots \quad \text{etc.} \quad \text{BVP solution strategies are discussed in the following sets of slides.}$$

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Eigenvalue BVP and vibration modes

Field equations are *ordinary differential equations*:

$$(EI_F(x) W''(x))'' - (T(x) W'(x))' - \omega_F^2 m(x) W(x) = 0 \quad W(x), \omega_F = ?$$

$$(EI_L(x) V''(x))'' - (T(x) V'(x))' - (\omega_L^2 + \Omega^2) m(x) V(x) = 0 \quad V(x), \omega_L = ?$$

Boundary conditions involve: $W, W', M(W''), S(W''')$

$$V, V', M(V''), S(V''')$$

Blade domain: $e_F R < x < R$

$$e_L R < x < R$$

BVP = ODE(s) + BC(s). Each BVP has an *infinite number* of solution pairs. Each solution pair constitutes a **mode of vibration** – an inherent dynamic characteristic the system under investigation. Therefore:

- Flapping: $W(x)$ (eigenfunction or mode shape) and ω_F^2 (eigenvalue).
- Lead-lag: $V(x)$ (eigenfunction or mode shape) and ω_L^2 (eigenvalue).

Eigenfunctions represent mode shapes (or free vibration shapes) and eigenvalues are related to the frequency of free (natural) vibration. In practice, only a first few solutions (e.g., three to ten) are of practical interest in DRS.

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Summary

- IV-BVPs can be transformed to the corresponding static, harmonic or eigenvalue BVPs.
- Static and harmonic BVPs search for *unique* solutions (deformations or amplitudes of vibration) for given loads.
- Eigenvalue BVPs offer an *infinite* number of solution pairs (modes of vibration) which are associated with free vibrations.
 - Modes of vibration are the system's dynamic characteristics with broad range of uses in dynamic design and analysis.
 - Modes of vibration form the constituents of free as well as forced vibrations – hence, they are the “building blocks” of any dynamic response.

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